

WESTERN SYDNEY
UNIVERSITY



Module 1

Creating a MATLAB Script

Editor & m-files

- The MATLAB script file has the extension **.m**
- MATLAB live scripts have the file extension **.mlx**
- The **.m** script file is also known as an **m-file**
- The script files in MATLAB are:
 - analogous to a txt-file in Notepad (**.m**) or a word processor document (**.mlx**)

Editor & m-files

- By using an **m-file** you can:
 - Save your list of executable commands (your program) in a sequential order
 - You can reopen and modify your program
 - They are useful for “debugging” (correcting errors)

Saving code in an m-file

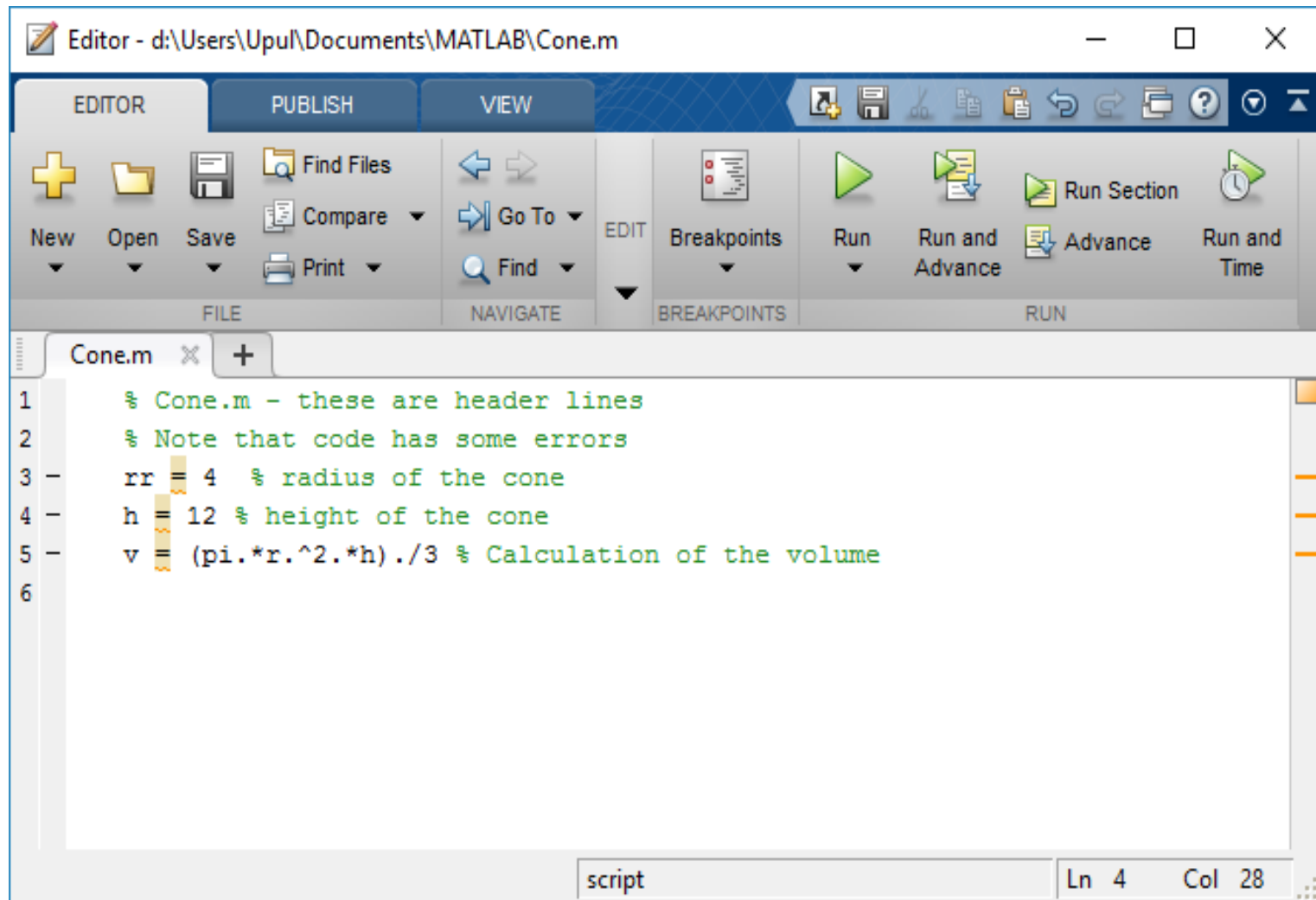
- Open the editor by:
 - Click the **New Script** icon located at the left most side of the toolbar.
 - or entering the command **edit** in the command window.
- Now enter the following lines of code to find the volume of a cone:

```
rr = 4  
h = 12  
v = (pi.*r.^2.*h) ./3
```
- After you have typed in the code, save it as cone.m (no need to physically type '.m')

Naming rules for MATLAB scripts

- Follow the same rules as naming a variable i.e.
 - Must begin with a LETTER
 - May only contain letters, numbers and underscores (_)
 - No spaces or punctuation marks allowed!

This is cone.m in the editor



comments were added, i.e. the words in green as a way to label the script

Executing a MATLAB script

- Now execute (or run) the program by pushing F5, or by typing on the command line

`>> cone` **<Enter>**

or by clicking the run button.

(Note that the **run button** looks like a green arrowhead pointing to the right in the Editor toolbar.)

- **Clicking run will run the program in sequential order starting from 'Line 1' up to the last line**
- **If you entered the code as written on the previous slide you will get an error!**

Executing a MATLAB script

- What went wrong?
 - Repair your program (change `rr = 4` to `r = 4`), save it, and run it again.
 - Now change the height to 24, save and run your program again (clicking the run button saves and executes your code).